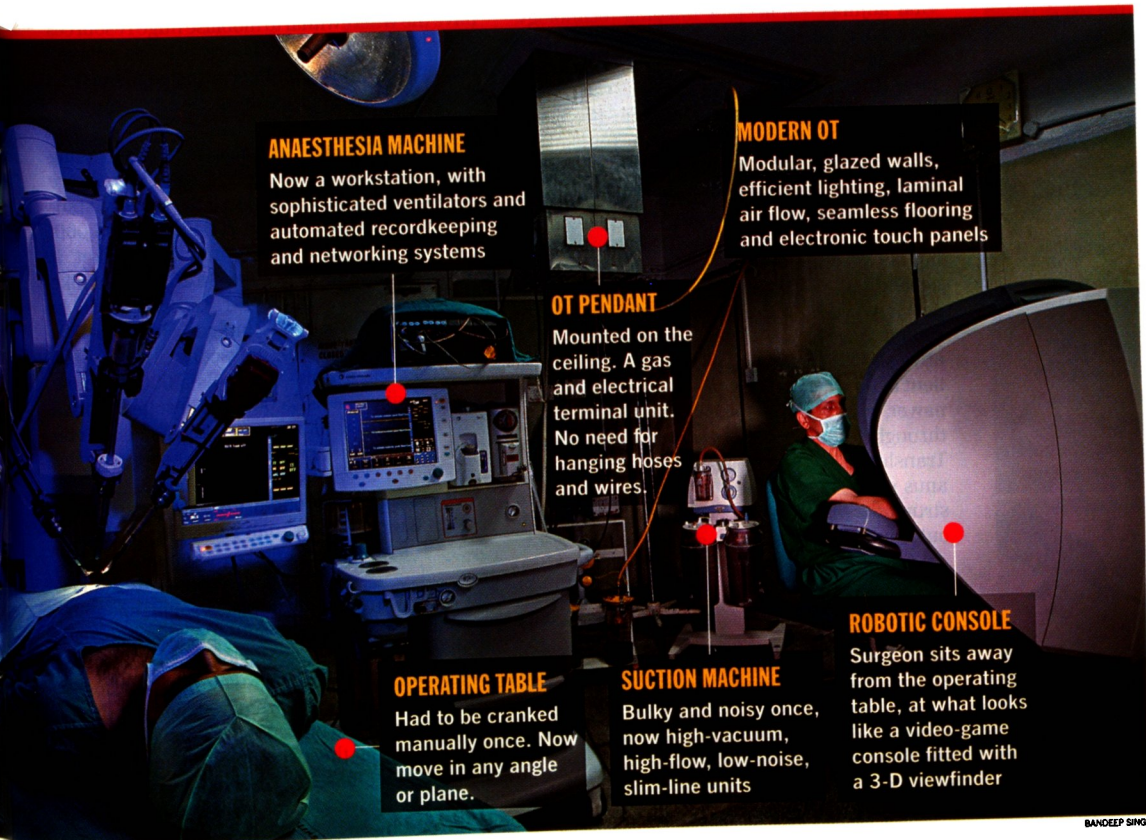


Forty years ago, when Trehan was a student, most devices in the operating room (OR) were rigid, allowing a few relatively simple procedures to be carried out with safety. Even in the '80s, gadgets were cumbersome and bulky, the operating table had to be cranked up or down with a jack and the overhead light fixed manually. Today, the OR looks more like a space capsule. Press a button and the table moves, in any angle and on any plane. Overhead lights are now aerodynamically controlled, with satellites lighting up hidden recesses of a patient's body. The crude electrocautery machine has given way to electrothermal Vessel Sealing Systems to control bleeding. From suturing, surgeons have moved on to staplers, while waterproof, biodegradable tissue adhesives and laser-assisted soldering are entering the OR. The old anaesthesia unit is now a "workstation", with sophisticated ventilators and automated record-keeping. "Twenty years ago, you had to go to the X-ray department, check out the angiograms, the blood report and all that. Today all is available on screen, allowing one to monitor those constantly," says Trehan. Not just that, the brick-and-mortar rooms are giving way to modular chambers, with fewer sharp edges to deter germ accumulation.

Equipment often hang from the ceiling for utmost hygiene. "Today's ORs are 20-25 per cent bigger," says neurosurgeon, Dr Sharad S. Kale of AIMS. The state-of-the-art OR under construction at AIMS will not only be enormous, it will have four satellite rooms to house all the new technology.

The glut of new technology corresponds to the "third revolution" in modern surgery (the first two came with antiseptics and anaesthesia). "At one time, the saying was, 'Big surgeon, big incision'," says Kumar, "now it's 'Big surgeon, small incision.'" Traditional surgery involved incisions large enough for the surgeon to put both his hands inside the patient's body. Since the '90s minimally invasive surgery (MIS) allowed surgeons to work through small punctures (or "keyholes") using chopstick-like tools teamed with a tiny camera. Instead of looking down at the patient, the surgeon looks up at enhanced images thrown up on a TV monitor. But "keyhole" surgery is counterintuitive: to move an instrument's tip to the left, surgeons have to push the handle to the right and vice versa. The 2-D monitor compromises depth, and the tools take away a lot of the touch sensation. Robotics, the next stage in surgery, takes the tools out of a surgeon's hands, performs precise move-



ANAESTHESIA MACHINE

Now a workstation, with sophisticated ventilators and automated recordkeeping and networking systems

MODERN OT

Modular, glazed walls, efficient lighting, laminar air flow, seamless flooring and electronic touch panels

OT PENDANT

Mounted on the ceiling. A gas and electrical terminal unit. No need for hanging hoses and wires.

OPERATING TABLE

Had to be cranked manually once. Now move in any angle or plane.

SUCTION MACHINE

Bulky and noisy once, now high-vacuum, high-flow, low-noise, slim-line units

ROBOTIC CONSOLE

Surgeon sits away from the operating table, at what looks like a video-game console fitted with a 3-D viewfinder

BANDEEP SINGH